

Critical Care Case Series 4: Post Extubation Stridor

Section 1: Case Summary

Scenario Title:	Post-extubation Stridor
Keywords:	Post-extubation edema, respiratory failure, stridor, adverse event
Brief Description of Case:	A 59 yo male was the intoxicated driver in a car vs pole MVC. He was intubated and admitted to ICU with TBI and ortho injuries. The patient has been agitated and delirious throughout his stay. The decision is made to extubate. However, the patient is quite agitated and requires sedation by nursing staff. Following extubation, he becomes stridorous and develops progressive respiratory failure. The team will need to manage the patient's airway, eventually requiring re-intubation. Subsequently, the patient's spouse will need to be informed of the patient's status as well as the indications for re-intubation.

Goals and Objectives	
Educational Goal:	Solidify approach to post-extubation stridor
Objectives: (Medical and CRM)	Medical: 1. Consider the differential for post-extubation respiratory failure 2. Identify management steps for post-extubation laryngeal edema 3. Consider the pros/cons of re-intubation following a failed extubation CRM: 1. Lead a multidisciplinary team during resuscitation 2. Convey management priorities to achieve a shared mental model 3. Value collective team knowledge and expertise in decision-making 4. Update family members following unexpected adverse event
EPAs Assessed:	Core EPA: C1 Resuscitating and coordinating care for critically ill patients Core EPA: C3 Providing airway management and ventilation

Learners, Setting and Personnel					
Target Learners:	☐ Junior Learners		☒ Senior Learners		☐ Staff
	☒ Physicians	☒ Nurses	☒ RTs	☐ Inter-professional	
	☐ Other Learners:				
Location:	☒ Sim Lab		☐ In Situ		☐ Other:
Recommended Number of Facilitators:	Instructors: one				
	Sim Actors: Bedside RN, RT, spouse				
	Sim Techs: 1				

Scenario Development	
Date of Development:	July 25 2017
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Critical Care Case Series 4: Post Extubation Stridor

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Edward Lumfield,			Age: 59	Gender: M	Weight: 90 kg
Presenting complaint: Major blunt polytrauma MVC					
Temp: 38	HR: 125	BP: 165/90	RR: 28	O ₂ Sat: 93%	FiO ₂ : 50%
Cap glucose: 5			GCS: 12(sedated)		
59 yo intoxicated driver in single-vehicle MVC, car vs pole					
Problem list:					
1) Mild TBI - small SDH, SAH (conservative management)					
2) Left rib fractures - small left pneumothorax (no drain)					
3) Left open ankle fracture – POD#1 ORIF					
4) C-spine –radiologically cleared but not cleared clinically					
5) On-going agitation and delirium					
Allergies: None					
Past Medical History:			Current Medications:		
Alcohol use disorder			Haldol 5mg IV prn		
			Lorazepam 2mg IV prn		
			Seroquel 50mg QID		

Section 2B: Extra Patient Information

A. Further History	
<i>This information will be provided by the nurse</i>	
59 yo male with AUD was brought in via EMS four days ago following a car vs pole MVC. He was intubated and admitted to ICU. He has been agitated throughout his stay.	
Injuries include a left ankle fracture that has been surgically repaired; as well as mild TBI, left rib fractures, and a small associated pneumothorax - all of which are being managed conservatively. His c-spine has been radiologically but not yet clinically cleared.	
30 minutes ago, he was extubated while being agitated.	
Due to agitation, the patient has received 5mg of Haldol IV and 2mg of Lorazepam IV to assist the nurses in suctioning his secretions post-extubation. He is now mildly sedated.	
The resident is called to the bedside because the patient has developed progressive stridor post-extubation.	
B. Physical Exam	
Cardio: tachycardic, regular	Neuro: moaning and still agitated at times, not answering questions, moving all 4 limbs, opens eyes to voice at times, PERLB
Resp: audible stridor, mild use of accessory muscles	Head & Neck: in c-spine collar
Abdo: unremarkable	MSK/skin: left ankle cast
Other: none	



Critical Care Case Series 4: Post Extubation Stridor

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>Adult male</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
On mannequin at SIM start: Arterial line ECG leads NIBP Pulse oximeter Cast on left ankle C-spine collar Venturi mask Available: Airway cart- including glidescope, bougie, LMA
C. Required Medications
Nebulized EPI IV fluids IV phenylephrine IV vasopressor infusion Induction medications Haldol IV Lorazepam IV
D. Moulage
Cast on left ankle C-spine collar
E. Monitors at Case Onset
☒ Patient on monitor with vitals displayed Monitors include: NIBP, ECG leads, RR, pulse oximeter, temp, and art line trace ☐ Patient not yet on monitor
F. Patient Reactions and Exam
Airway- audible stridor Breathing- mild use of accessory muscles, equal BS bilaterally Circulation – skin appears well perfused Disability – moving all 4 limbs, eyes open occasionally to voice, PERL, pt is moaning and not answering questions, becomes intermittently agitated

Critical Care Case Series 4: Post Extubation Stridor

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Bedside RN	- You are in the room when the resident arrives. - The patient has developed progressive stridor since extubation, and you have called back the trainee to the bedside. - You update the trainee on the patient's condition and answer questions based on the additional info provided above. - You are not confident that this extubation will be successful and are increasingly worried about the patient as the scenario progresses, especially if the trainee hesitates to re-intubate. You are collaborative, but anxious. - You argue in favor of reintubation with increasing insistence as patient's vitals change - Post-intubation, you suggest the trainee call in the spouse (SDM) as they have been waiting outside for an hour to visit and need an update
RT	- You are in the room when the trainee arrives. - You suction the patient each time he desats, which brings the sat up for a few minutes - You think the patient should be re-intubated as soon as possible and would be more comfortable if the staff was at the bedside as you don't know this trainee very well. You object strongly if suggestion of NIV is made as an initial approach.
Spouse/SDM	- Your spouse was admitted to the hospital four days ago after a car accident. - The police said he was intoxicated with alcohol at the time. - You feel guilty for having "let him" drive intoxicated in the past, even if you had mentioned to him dozen of times in the past how dangerous that was. - You've felt quite overwhelmed since the accident. Your general sense is that he is getting better as the doctors were talking about removing the tube in his mouth yesterday. - Having said that, you don't understand why he's so agitated and confused. He's usually a very quiet man. - You have arrived at the hospital an hour ago for your daily visit, but the RN has been asking to wait outside as they were busy. You are starting to worry that something is wrong. - When the trainee comes to speak with you, make clear your frustration that you do not know what is going on. You were not aware that the tube was being removed today (although it was mentioned as a possible next step yesterday). You want to know why they had to put the tube back in and what this means for your spouse's recovery.
Possible roles on standby	Charge RN, Junior resident, colleague ICU resident/fellow



Critical Care Case Series 4: Post Extubation Stridor

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: SR HR: 125/min BP: 165/90 RR: 28/min O ₂ SAT: 93% (on 50%) T: 38°C GCS: 12	Mildly sedated and does not answer questions, moaning. Occasionally, becomes more agitated. Has audible stridor and mild accessory muscle use	<u>Expected Learner Actions</u> ☐ Receive bedside report ☐ Review chart ☐ Physical exam ☐ Suction	<u>Modifiers</u> - Sats increase to 95% temporarily with suctioning <u>Triggers</u> - After 3 mins, move to next phase	Each time the RT suction the patient, sats temporarily improve
2. At 3 min HR: 127/min BP: 160/90 RR: 28/min O ₂ SAT: 75% (on 50%)	Patient desats to 75%	<u>Expected Learner Actions</u> ☐ Continue suction ☐ FiO ₂ up to 100% ☐ BMV ☐ Consider re-intubation	<u>Modifiers</u> - O ₂ Sats slow increase with each intervention <u>Triggers</u> - After 2 more mins or decision to re-intubate, move to the next phase	RN and/or RT strongly suggest reintubation
3. At 5 mins HR: 127/min BP: 160/90 RR: 30/min O ₂ Sat: 92% (on 100%)	Stridor becomes louder RR increases to 30/min Increased agitation	<u>Expected Learner Actions</u> ☐ Nebulized Epi +/- corticosteroids ☐ Consider NIPPV ☐ Decision to re-intubate	<u>Modifiers</u> - Temporary improvement in stridor with epi neb then remains stridorous until intubation - Pt is agitated with no change in VS <u>Triggers</u> - Begin re-intubation, move to next phase	If NIV is suggested RT will disagree in favour of ET May prompt ET if necessary to move on to next stage Note consideration of c-spine in ETI
4. ETI HR: 127/min BP: 80/50 RR: 30/min O ₂ SAT: 96%	BP drops with induction Sats improve post-ETT	<u>Expected Learner Actions</u> ☐ Intubation ☐ IVF bolus ☐ IV phenylephrine ☐ IV vasopressor infusion ☐ Restart sedation	<u>Modifiers</u> - BP will normalize with pressure support <u>Triggers</u> - BP managed, move on to discussion with spouse	Once this stage is complete, RN to suggest that trainee speak with spouse



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Appendix A: Laboratory Results

<u>CBC</u> WBC 12 Hgb 90 Plt 250 <u>Lytes</u> Na 137 K 4.0 Cl 101 HCO ₃ <u>VBG</u> pH pCO ₂ pO ₂ HCO ₃ Lactate	<u>Cardiac/Coags</u> Trop D-dimer INR aPTT <u>Biliary</u> AST ALT GGT ALP Bili Lipase <u>Tox</u> EtOH ASA Tylenol Dig level Osmols <u>Other</u> B-HCG
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Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Progress Note

Date: This AM

MVC Day#4

59yo, intoxicated driver, Car vs. Pole

Initial GCS M4, E2, V1

Injuries:

- 1) Mild TBI - small SDH, SAH (conservative management)
- 2) Left rib fractures - small left pneumothorax (no drain)
- 3) Left open ankle fracture post-ORIF D#3

Neuro: Agitated - answers simple questions

CV: stable

Resp:

- min settings
- secretions amount moderate - strong cough
- cuff leak +ve
- xray yesterday - atelectasis LLL - no visible pneumotx

GI: feeds on hold for ext., No BM yet

UO: ok - neg 1L - lytes ok

Heme: Hg stable - WBC 12' (10' yest.)

ID: Tmax 38.1

Active issues:

1. Delirium

- increase seroquel to 50mg QID
- PRN haldol/benzos

2. Ventilation

- plan to extubate today

3. C-spine clearance

- waiting to be cleared clinically - pt not collaborative enough thus far (no fracture on CT)

Critical Care Case Series 4: Post Extubation Stridor

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

1- Leadership (ref: CRM Royal College - Chapter on Leadership)

- It is important for trainees to strike the right balance between assertiveness & collaborative approach
- Discuss trainee's ability to communicate plan and organize resources in a timely manner
- Discuss their approach to considering suggestions from other team members

2- Family update

- Assess clarity of the explanations provided
- Discuss trainee's ability to convey support/reassurance
- Ensure lack of defensiveness in difficult conversations

3- Management stridor post-extubation (Plumijms et al, 2015; StatPearls, 2021)

- Assess soundness of trainee's initial approach/back up plan.
- Examine their rationale behind decision-making.
- Discuss differential for post-extubation respiratory failure (PRF):
 - PRF occurs in up to 10% of extubations and can be caused by inability to ventilate spontaneously or by inability to tolerate removal of ETT (extubation failure). Causes of extubation failure include upper airway obstruction and secretions.
- Discuss the etiology of post-extubation laryngeal edema (PLE) and stridor:
 - Post-extubation laryngeal edema (PLE) is caused by trauma to and pressure exerted on the larynx during intubation.
 - While many patients may have some degree of mild laryngeal edema following intubation, most are asymptomatic or mildly symptomatic from it. Stridor indicates a more severe obstruction.
 - In severe cases, PLE/S can lead to respiratory failure and require re-intubation.
- Discuss identification of patients at high risk for PLE/stridor:
 - Risk factors for PLE are: female gender, large tube size, prolonged intubation, agitation, high cuff pressures.
 - Its presence is often not clinically apparent until after extubation.
 - There is no reliable test to identify patients at high risk for PLE, though cuff leak test or use of ultrasound may identify low risk patients.
- Discuss the management of post-extubation stridor:
 - Management includes nebulized epi and IV or nebulized corticosteroids, though evidence is not robust.
 - If stable but not improving, patients should be monitored for 1 hour prior to reintubation.
 - In the case of respiratory failure, reintubation should not be delayed.
 - Use of NIV is not indicated since it does not improve resp status and delays intubation.

References

1. Duggan LV. Optimizing Crisis Resource Management to Improve Patient Safety and Team Performance-A handbook for acute care health professionals. Royal College of Physicians and Surgeons of Canada, 2017.
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